

WHAT IS A DINOSAUR?

Dinosaurs are reptiles that evolved early in the Mesozoic Era, a period of time spanning from 252 to 66 million years ago. They belong to a larger group of animals called archosaurs, which also includes the crocodylians. All dinosaurs are vertebrates (animals with backbones), but they also share certain characteristics that are not seen in other animal groups. These unique traits help paleontologists distinguish dinosaur fossils from those of other prehistoric animals. If an excavation reveals a bone or a partial skeleton that has one or more of the traits shown on these pages, it is very likely to be a dinosaur fossil.

FEATHERS

Recent research suggests that dinosaurs evolved feathers well before the appearance of the first birds. Many dinosaurs probably had feathers on their body, while some—such as this *Archaeopteryx*—even had feathered wings, which confirmed that birds evolved from dinosaurs.



Most dinosaurs had a strong, muscular tail that helped them balance their weight while moving.



SKIN TEXTURE

The skin of many dinosaurs was covered in small scales, some of which have been found preserved alongside skeletons, as was the case with this hadrosaur skin fossil. Dinosaur skin was waterproof and the scales probably provided some protection against insect bites.

Herbivores used their long gut to digest tough plant matter.

Dinosaurs had flexible, hingelike ankles that helped them move fast.

